

Renzo Martin Mendoza Rabanal

"Engineering and control for the real world"

Hi, I am a BSc graduate in Mechatronics Engineering from the Pontificia Universidad Catolica del Peru (PUCP). I am currently a research engineer at the  **Hernandez-Nunez Lab** [↗](#). I completed my in-person appointment at Harvard University, but I will continue our projects at Stanford University starting in 2026!

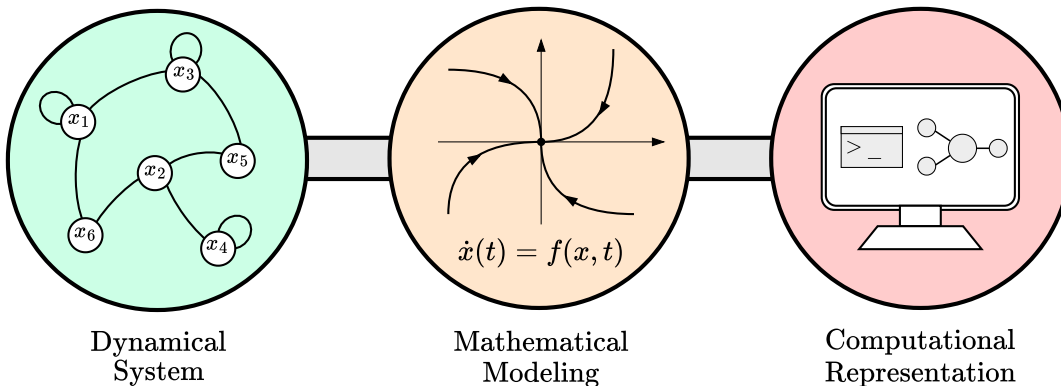
I am passionate about control theory and its applications in real-world systems. Specifically, I am passionate about using nonlinear system analysis to understand the dynamics of complex systems and unravel the principles underlying their fundamental behaviors.

This portfolio showcases projects I have been involved in, along with descriptions and insights into each experience. You can also view a summarized online version on my [Website](#) [↗](#). Finally, a visual representation of the areas and topics to which I am dedicated is shown below.



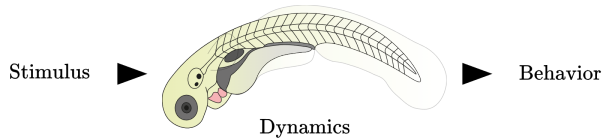
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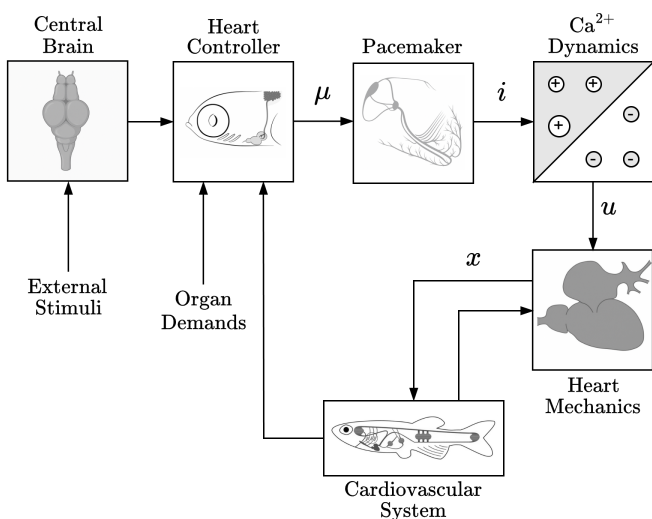


Research Spotlight

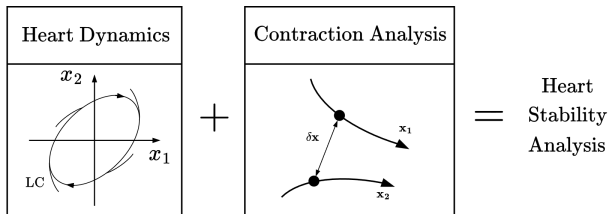
Heart dynamics *in vivo*



Brain-Heart feedback



Is the heart stable?



Zebrafish: Behavior and Modeling

May 2024 - Present

We are currently investigating the organ dynamics of larval zebrafish. We will use control theory and system identification approaches to describe the brain-heart interactions. This project is still in execution, so more details will be released soon!

We are also beginning a new focus on analyzing zebrafish heart stability and hope to demonstrate that the heart is a robust system, leveraging these findings for future applications.

Contribution and technical details

On this project, I am participating in computational tasks that involve heart data pre-processing and tracking using deep learning models, algorithm development for metric extraction, and behavioral clustering using data-driven approaches. Additionally, I implemented a motion correction algorithm for calcium imaging videos from a paper that was not available for MATLAB. I shared my code on the MATLAB File Exchange forums for use by other community members.

Additionally, I proposed analyzing heart dynamics and stability using a control theory approach called Contraction Analysis. I am currently creating scripts to solve a partial differential equation, with the goal of identifying a contraction region where heart dynamics are exponentially stable.

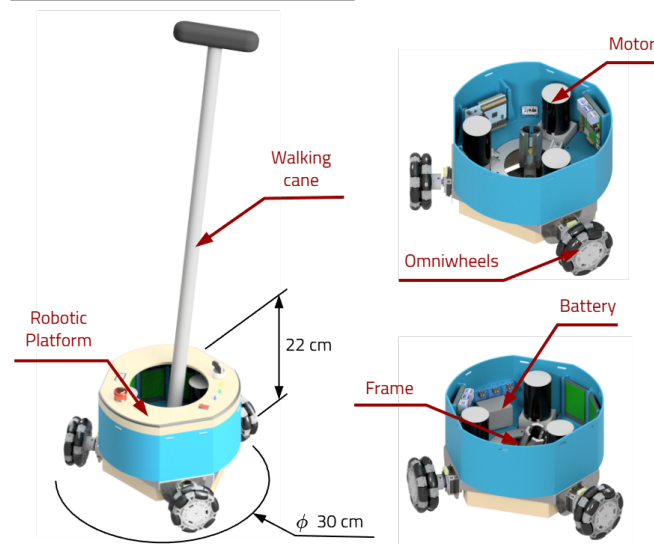
Skills

- Signal and image processing
- Data-driven clustering
- System identification
- Nonlinear system analysis

Resources

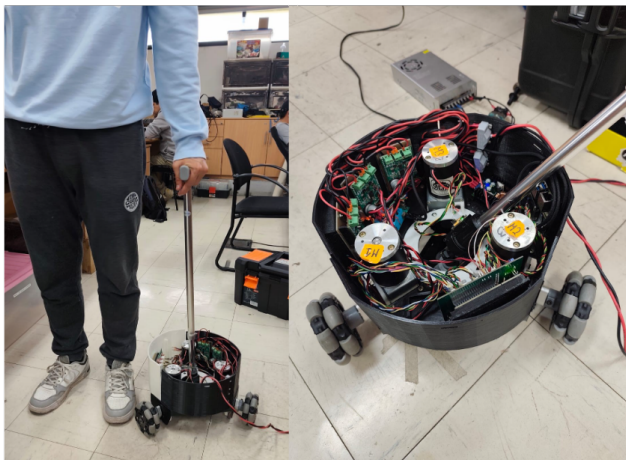
- [MATLAB repository](#)

Conception



(*) Design credit: Alain Chupa

Robotic Platform Tests



Robotic Cane for Walking Assistance

July 2024 - Present

Our team is developing a modular robotic platform designed to be attached to a walking cane to assist people with disabilities in walking and maintaining balance, thereby making their daily lives easier. This mobile platform is designed to displace and rotate without restrictions using an omnidirectional array of wheels.

We are finishing the system implementation and starting testing in a controlled environment. Additionally, we will initiate the process to patent our design!

Contribution and technical details

I had the opportunity to be part of this project from the beginning. I was part of the development of the idea, methodology, and proposal that successfully secured a grant from PUCP. My initial focus was on control and system analysis, where I mathematically modeled the system as a spherical inverted pendulum, taking into account uncertainties and disturbances within a Lagrangian framework. Additionally, I explored various controllers based on linear matrix inequalities to stabilize the system.

Since the early stages of the mechatronic design, I took on an active and leadership role, collaborating closely with team members. I supported the mechanical, electronic, and algorithmic aspects of the project, working along with an undergraduate student. My main contribution was in the algorithmic design, where I drew on my previous experience with estimating the system dynamics and embedding a low-level controller for DC motors, as well as in developing kinematic tracking control for the omnidirectional platform.

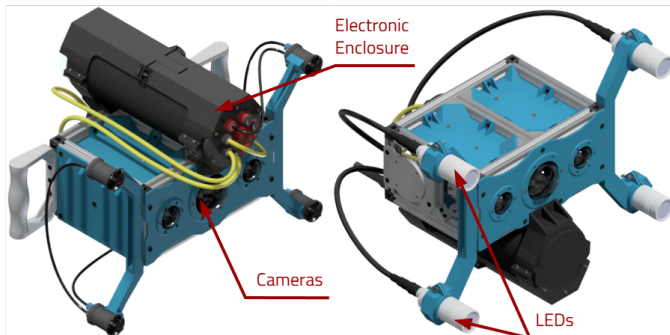
Skills

- Research project formulation
- Mechatronic design
- Control theory
- Linear system identification

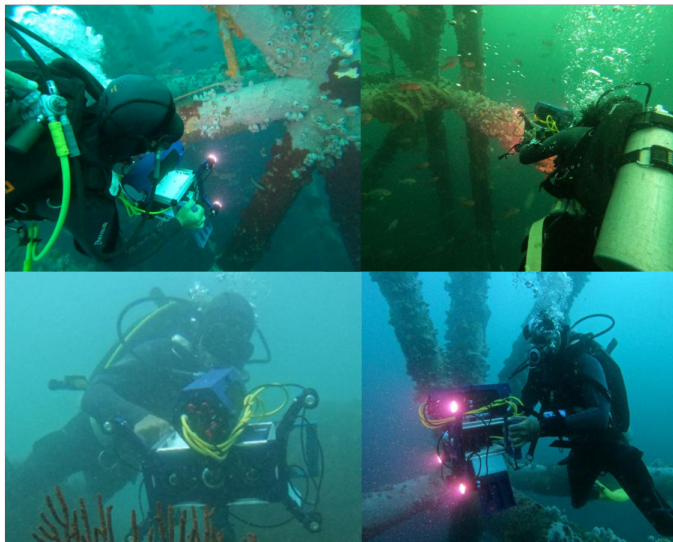
Resources

- [Photos and videos](#)

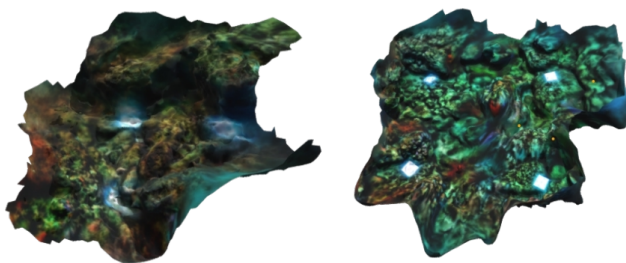
Design



Implementation



Results



(*) 3D reconstruction courtesy of Daniel Menacho

Underwater Image Acquisition System

January 2021 - January 2023

During this project, our team developed an image acquisition system for underwater environments and a photogrammetry pipeline to obtain the necessary data for digitally reconstructing those environments. The results of this project intend to reduce the pollution generated by traditional tourist activities and promote eco-tourism.

The system was tested in real-world environments along the Peruvian coast, including [Pucusana](#) and [Los Organos](#), and showed that the device performs well in subaquatic and low-light conditions.

Contribution and technical details

The entire development of the underwater acquisition system was the focus of my undergraduate thesis. During this project, I developed the mechanical, electronic, and algorithmic designs of the system. The mechanical enclosure was built to withstand pressures at depths of up to 200 meters. I incorporated a single-board computer and a microcontroller to ensure the electronic subsystem performed optimally. Additionally, I shared my designs in an online repository for the community to access.

I am personally proud of my proposal for a control algorithm that regulates the underwater LEDs, which provide illumination in low-light conditions. This approach combined image processing and control theory concepts to improve the brightness quality of recorded images while reducing the system's power consumption.

Skills

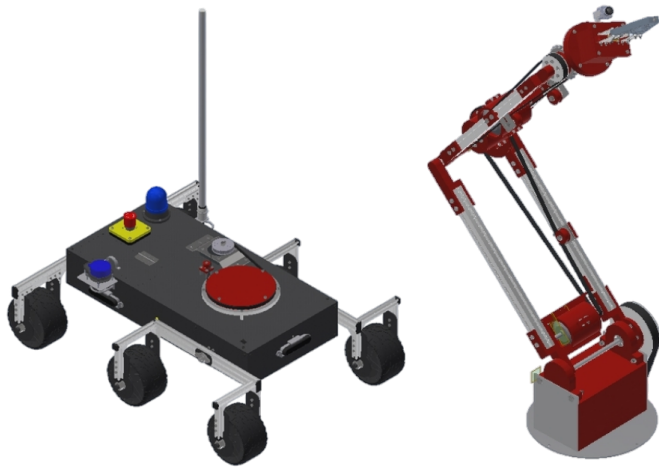
- Mechatronic Design
- Systems integration & programming
- Control theory
- Real-world deployment of systems

Resources

- [Photos and videos](#)
- [System 3D models](#)
- [Underwater environments](#) (by Daniel Menacho)

Personal Projects

Initial Designs



Orion Team and ERC



Rover Project and Orion Team

January 2021 - January 2023

This project was an initiative by a group of undergraduate students, including myself, who aimed to compete in international events and showcase our university, as there was no precedent for such an effort prior to this. We started developing our initial design of an autonomous rover robot as a project for a course called Mechatronic Design Methodology.

After these initial endeavors, we recruited additional members, and our team was subsequently named the Orion Rover Team. We successfully competed in the remote edition of the 2023 European Rover Challenge (ERC). I am proud to emphasize that we achieved the highest ranking during the classification stage, and we were the first Peruvian team to participate in this international competition. During this competition, we developed solutions for aerospace maintenance tasks, controlling a robotic arm and a small rover.

Contribution and technical details

Initially, I participated in the mechanical design of the robotic arm attached to the rover. After that, I focused on implementing the robot model virtual simulation environments using ROS and Gazebo.

To compete in the ERC 2023 remote edition, I served as the technical leader, overseeing team members and assigning tasks to ensure success. Additionally, I actively contributed by developing scripts to control a UR3 robotic arm using CHOMP and STOMP trajectory planners within the MoveIt framework.

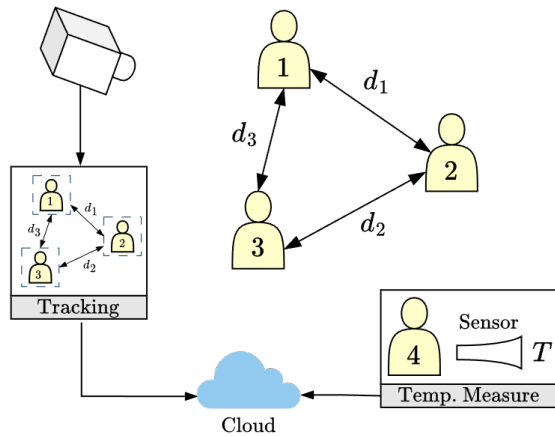
Skills

- Mechanical design
- ROS, Moveit and robot simulation
- Robotic arm control programming

Resources

-  [Photos, videos and slides](#) 
-  [ERC](#) 
-  [Code](#) 

Conceptual Idea



Implementation and tests



Crowd Monitoring System

June 2020 - January 2021

In a course on object-oriented programming, our team proposed a project to monitor people's temperature and the number of individuals in an environment to help reduce the transmission of SARS-CoV-2. The system included a temperature registration module, a subsystem to estimate the distance between individuals, and real-time data updates in the cloud. We implemented a local version of the project and presented our results at the International Scientific Congress (ECI) in 2021, Peru.

Contribution and technical details

I was responsible for implementing and testing Haar cascade object detection algorithms, as well as programming the system's user interface.

Skills: Object-oriented programming, Object detection, Mechatronic systems

Resources

- [Photos](#)

Path Planning for Service Robots

January 2021 - March 2021

During an undergraduate course in autonomous vehicles, our team was responsible for implementing an autonomous service robot in simulation. We developed a solution in the ROS environment.

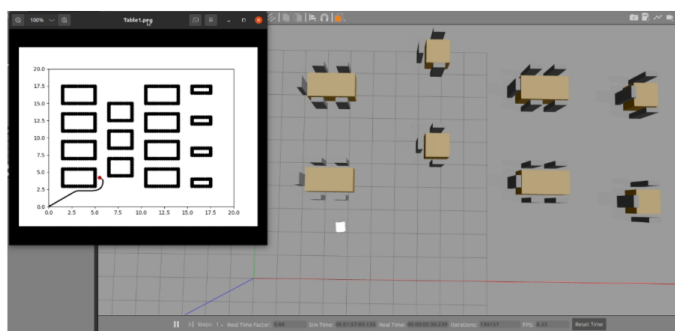
Contribution and technical details

I researched and implemented a potential field-based path planning algorithm for a service robot. Moreover, I successfully implemented kinematic tracking control for this robot in a simulation environment, which enabled the completion of point-to-point movements.

Skills: ROS and robot simulation, Path planning and tracking control

Resources

- [Figures and videos](#)



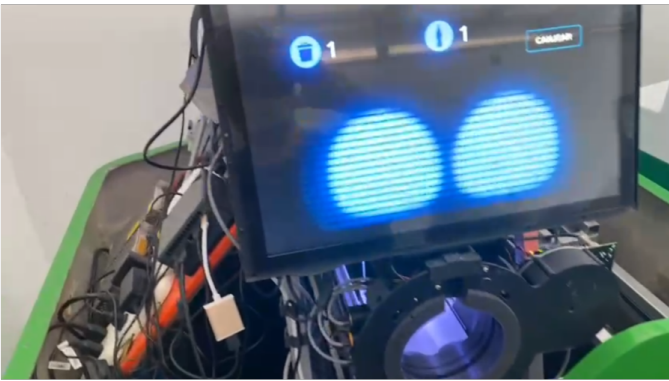
Industry Experience



Technical Trainer in Ferreyros

April 2022 - March 2023

As a technical trainer engineer, I enjoyed specializing in autonomous mining trucks and their electric powertrain systems. During this experience, I was passionate about developing projects that focused on teaching these concepts to the technical staff, as I truly value sharing my knowledge with people.



Engineer in Cirsys

March 2022 - April 2022

After graduating, I was approached by a former university instructor to join his startup, which developed the first intelligent recycling robot, IrBin, in Peru. I served as a junior engineer, repairing, enhancing, and testing the electrical and algorithmic systems of IrBin.

Expertise

- **Control Theory:** PID control, Kalman filter, linear matrix inequalities, linear parameter-varying control, nonlinear systems.
- **Robotics and Mechatronics:** robot kinematics and dynamics, autonomous systems, ROS.
- **Data-driven and image processing:** iterative denoising trees, proper orthogonal decomposition, particle image velocimetry.
- **Programming Languages:** MATLAB, Python, C, C++.
- **Languages:** english, spanish.

Awards, Honors and Funding

- **CAP 2024 Grant:** grant of 28,000 USD allocated by PUCP to develop the robotic cane for walking assistance.
- **BEA Scholarship:** awarded seven times to cover 70% of college tuition.
- **Medal and Honor Roll:** Recognized three times for academic excellence.